Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R12-E1_000
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)

Name	SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)
Description	
Analysis	Stress, strain/ simplified loading

Design

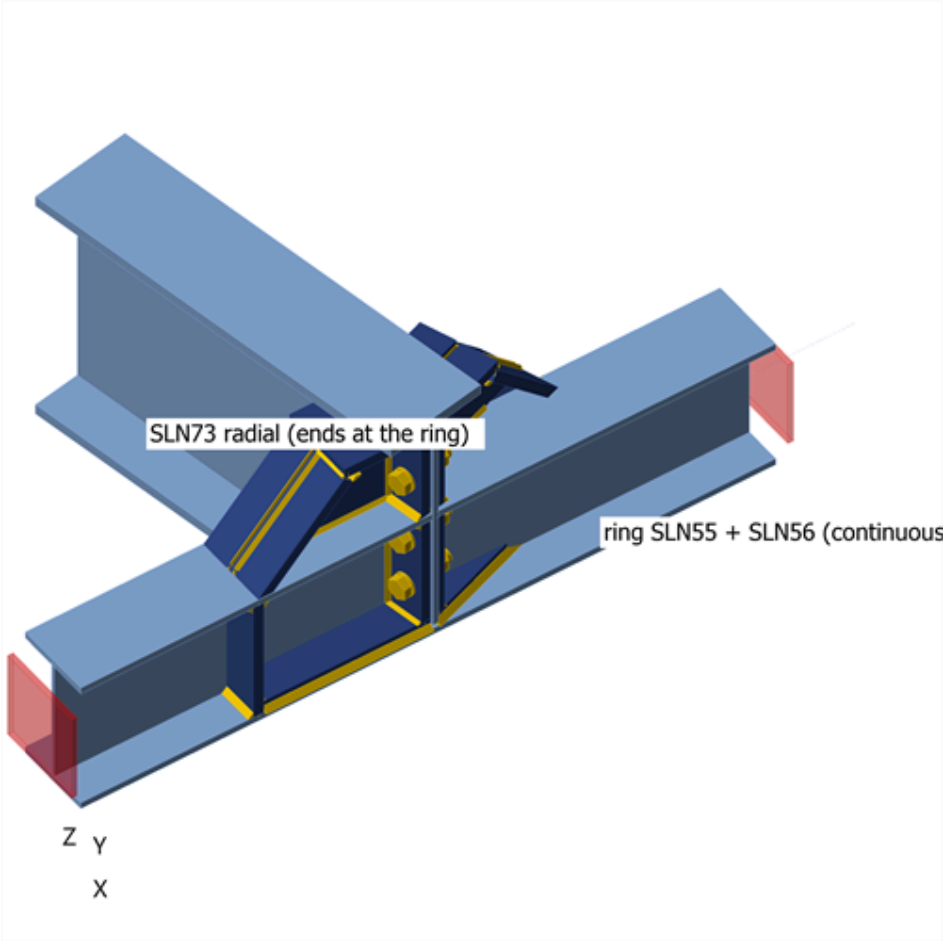
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN73 radial (ends at the ring)	1 - UB457x191x98	0.0	0.0	0.0	76	0	0
ring SLN55 + SLN56 (continuous)	2 - UB254x146x43	-100.0	0.0	0.0	0	0	-104

Supports and forces

Name	Support	Forces in	X [mm]
SLN73 radial (ends at the ring) / begin		Position	0
ring SLN55 + SLN56 (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
ring SLN55 + SLN56 (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0



SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous), Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M24 10.9	24	900.0	1000.0	452

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_102	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_103	SLN73 radial (ends at the ring) / Begin	9.6	-0.1	-86.2	-0.1	-45.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_104	SLN73 radial (ends at the ring) / Begin	7.9	0.0	-86.4	-0.1	-45.4	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_105	SLN73 radial (ends at the ring) / Begin	8.8	-0.1	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_106	SLN73 radial (ends at the ring) / Begin	8.0	0.0	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_107	SLN73 radial (ends at the ring) / Begin	1.8	-0.1	-7.8	0.0	-4.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_108	SLN73 radial (ends at the ring) / Begin	1.7	0.1	-8.4	0.0	-4.6	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_109	SLN73 radial (ends at the ring) / Begin	1.4	0.1	-8.1	0.0	-4.5	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_110	SLN73 radial (ends at the ring) / Begin	0.6	-0.1	-8.1	0.0	-4.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_111	SLN73 radial (ends at the ring) / Begin	7.4	-0.1	-61.7	0.0	-32.6	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_112	SLN73 radial (ends at the ring) / Begin	6.3	0.1	-62.3	0.0	-32.8	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_113	SLN73 radial (ends at the ring) / Begin	6.7	0.0	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_114	SLN73 radial (ends at the ring) / Begin	5.6	-0.1	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_115	SLN73 radial (ends at the ring) / Begin	2.4	-0.1	-13.1	0.0	-7.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_116	SLN73 radial (ends at the ring) / Begin	2.1	0.1	-13.6	0.0	-7.4	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_117	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-13.3	0.0	-7.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_118	SLN73 radial (ends at the ring) / Begin	1.1	-0.1	-13.3	0.0	-7.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_119	SLN73 radial (ends at the ring) / Begin	11.2	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_120	SLN73 radial (ends at the ring) / Begin	8.1	0.0	-89.8	-0.1	-47.2	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_121	SLN73 radial (ends at the ring) / Begin	8.5	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_122	SLN73 radial (ends at the ring) / Begin	7.6	0.0	-89.8	-0.1	-47.2	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_123	SLN73 radial (ends at the ring) / Begin	10.8	-0.1	-68.9	0.0	-36.3	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_124	SLN73 radial (ends at the ring) / Begin	6.7	0.1	-68.9	0.0	-36.3	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_125	SLN73 radial (ends at the ring) / Begin	6.2	0.0	-68.9	0.0	-36.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_126	SLN73 radial (ends at the ring) / Begin	4.8	-0.1	-68.9	0.0	-36.3	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_127	SLN73 radial (ends at the ring) / Begin	5.7	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_128	SLN73 radial (ends at the ring) / Begin	2.5	0.1	-20.2	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_129	SLN73 radial (ends at the ring) / Begin	1.5	0.0	-20.3	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_130	SLN73 radial (ends at the ring) / Begin	0.3	-0.1	-20.3	0.0	-10.9	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_135	SLN73 radial (ends at the ring) / Begin	7.7	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_141	SLN73 radial (ends at the ring) / Begin	2.3	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_142	SLN73 radial (ends at the ring) / Begin	1.8	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_143	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_144	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_145	SLN73 radial (ends at the ring) / Begin	10.1	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_146	SLN73 radial (ends at the ring) / Begin	8.3	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_147	SLN73 radial (ends at the ring) / Begin	9.4	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_148	SLN73 radial (ends at the ring) / Begin	9.0	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_151	SLN73 radial (ends at the ring) / Begin	6.9	-0.1	-46.4	0.0	-22.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_152	SLN73 radial (ends at the ring) / Begin	9.3	0.0	-63.5	0.0	-35.9	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_153	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.4	0.0	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_154	SLN73 radial (ends at the ring) / Begin	7.0	-0.1	-46.8	0.0	-22.5	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	7.1 > 5.0%	**Not OK!**
Bolts	59.1 < 100%	OK
Welds	102.8 > 100%	**Not OK!**
Buckling	Not calculated	

Plates

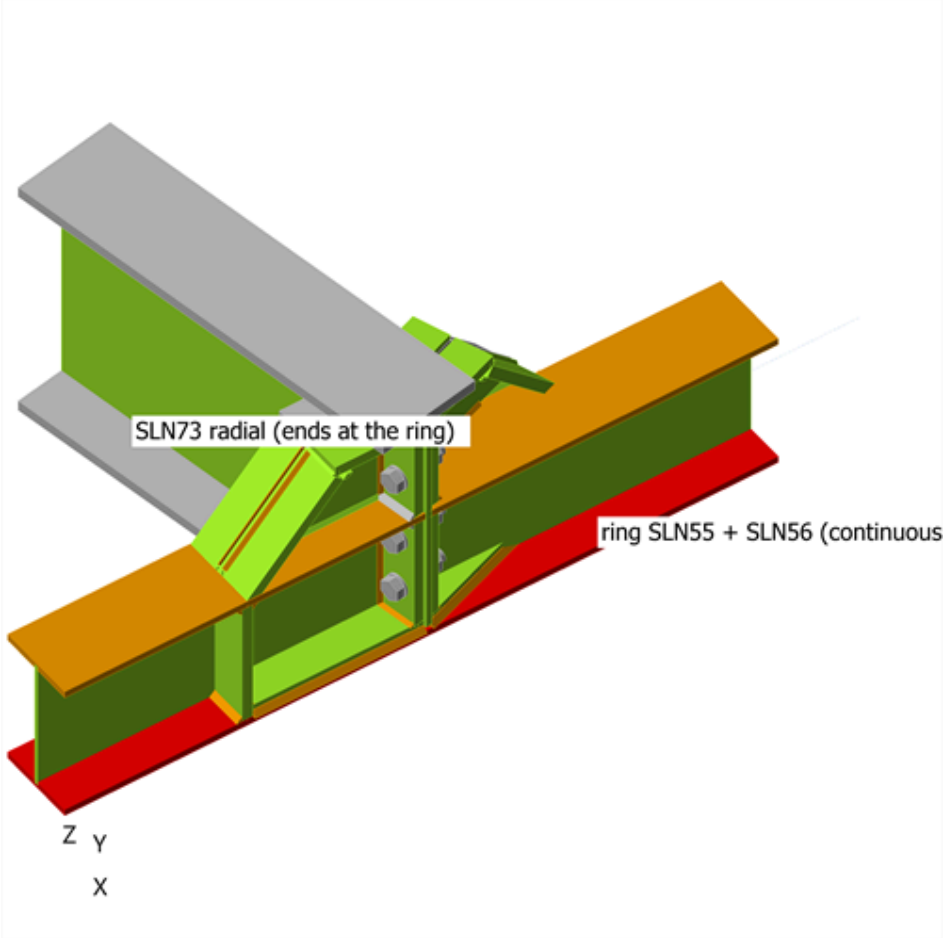
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN73 radial (ends at the ring)-bfl 1	19.6	C LC_120	18.0	0.0	0.0	OK
SLN73 radial (ends at the ring)-tfl 1	19.6	C LC_122	124.4	0.0	13.5	OK
SLN73 radial (ends at the ring)-w 1	11.4	C LC_122	263.8	0.0	71.7	OK
ring SLN55 + SLN56 (continuous)-bfl 1	12.7	C LC_120	290.0	7.1	0.0	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	12.7	C LC_135	283.2	3.9	0.0	OK
ring SLN55 + SLN56 (continuous)-w 1	7.2	C LC_122	275.9	0.4	0.0	OK
395x145x20 fin plate upper (SLN55 side)	20.0	C LC_120	275.3	0.1	610.7	OK
395x145x20 fin plate lower outer (SLN55 side)	20.0	C LC_120	255.1	0.0	138.1	OK
395x145x20 fin plate lower inner (SLN55 side)	20.0	C LC_119	275.6	0.3	68.7	OK
395x145x20 fin plate upper (SLN56 side)	20.0	C LC_122	258.2	0.0	613.2	OK
395x145x20 fin plate lower outer (SLN56 side)	20.0	C LC_119	253.4	0.0	136.3	OK
395x145x20 fin plate lower inner (SLN56 side)	20.0	C LC_119	275.1	0.1	55.1	OK
20 ring stiffener at 315 (SLN55 side)	20.0	C LC_122	276.9	0.9	0.0	OK
20 x 295 plate on ring bottom flange (SLN55 side)	20.0	C LC_122	275.7	0.3	0.0	OK
10 bracket web plate (SLN55 side)	10.0	C LC_122	275.3	0.1	0.0	OK
20 bracket top plate a (SLN55 side)	20.0	C LC_122	275.1	0.0	84.5	OK
20 inclined stiffener a (SLN55 side)	20.0	C LC_122	221.3	0.0	0.0	OK
20 bracket top plate b (SLN55 side)	20.0	C LC_122	275.5	0.2	66.5	OK
20 inclined stiffener b (SLN55 side)	20.0	C LC_122	275.6	0.3	0.0	OK
20 ring stiffener at 315 (SLN56 side)	20.0	C LC_122	275.7	0.3	0.0	OK
20 x 295 plate on ring bottom flange (SLN56 side)	20.0	C LC_119	269.0	0.1	0.0	OK
10 bracket web plate (SLN56 side)	10.0	C LC_135	275.0	0.0	0.0	OK
20 bracket top plate a (SLN56 side)	20.0	C LC_135	276.2	0.6	5.6	OK
20 inclined stiffener a (SLN56 side)	20.0	C LC_135	275.1	0.0	0.0	OK
20 bracket top plate b (SLN56 side)	20.0	C LC_122	253.8	0.0	146.4	OK
20 inclined stiffener b (SLN56 side)	20.0	C LC_135	275.1	0.0	0.0	OK

Design data

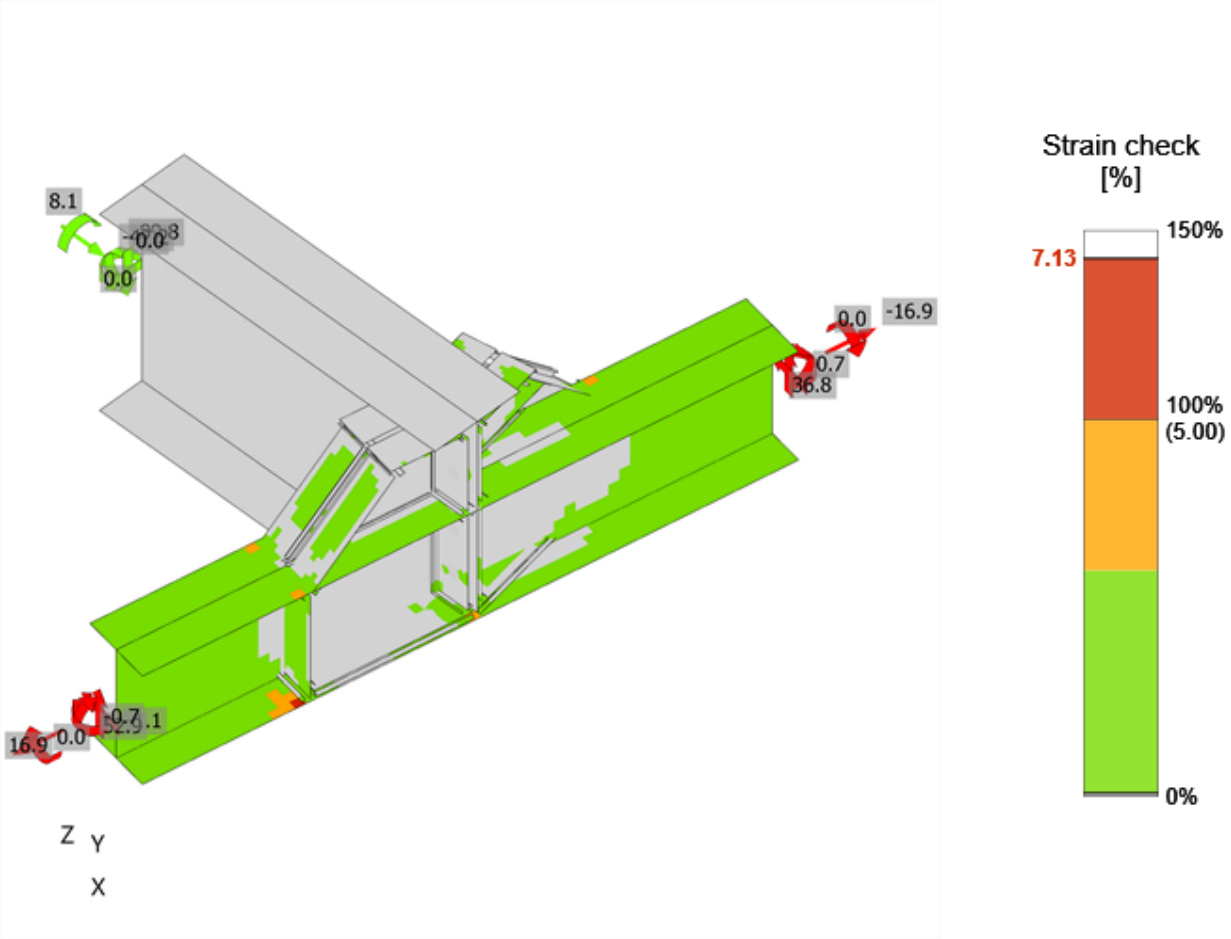
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

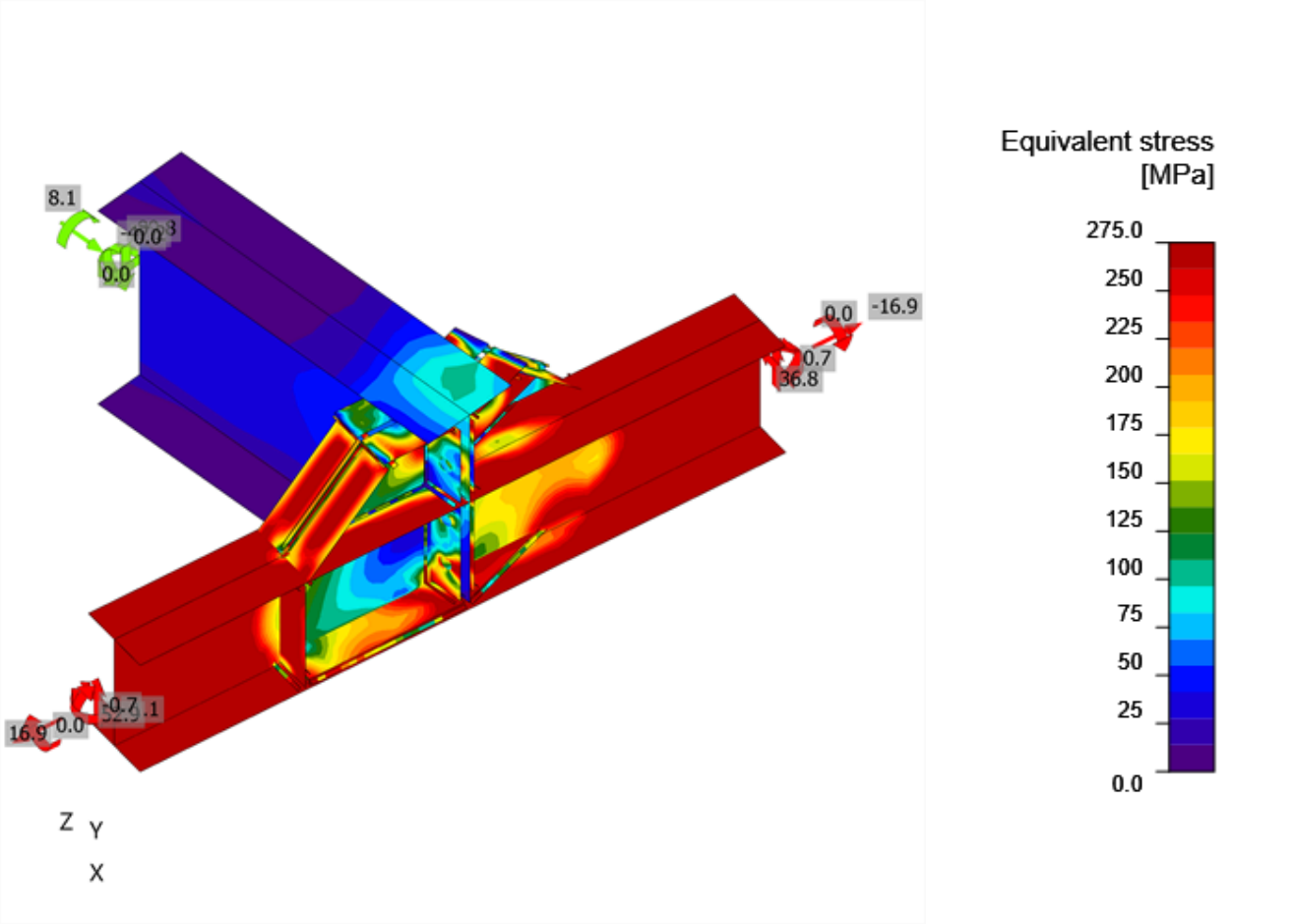
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

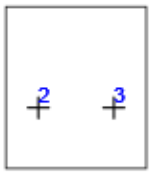
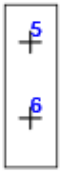
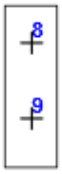


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M24 10.9 - 1	C LC_122	50.9	36.8	235.3	20.0	31.0	40.4	OK	OK
	B3	M24 10.9 - 1	C LC_122	13.0	36.9	119.3	5.1	59.1	29.8	OK	OK
	B5	M24 10.9 - 1	C LC_122	6.0	14.9	235.3	2.4	11.2	12.3	OK	OK
	B6	M24 10.9 - 1	C LC_153	2.1	39.7	235.3	0.8	33.0	28.7	OK	OK
	B8	M24 10.9 - 1	C LC_153	0.0	12.9	235.3	0.0	9.5	9.2	OK	OK
	B9	M24 10.9 - 1	C LC_119	51.4	35.8	235.3	20.2	29.8	39.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M24 10.9 - 1	254.2	492.7	141.2

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
10 bracket web plate (SLN55 side)	20 bracket top plate a (SLN55 side)	-	128	-	-	-	-	-	-	-	-	OK	OK
		-	128	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN55 side)	20 bracket top plate b (SLN55 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate a (SLN56 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate b (SLN56 side)	-	128	-	-	-	-	-	-	-	-	OK	OK
		-	128	-	-	-	-	-	-	-	-	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 4.2 ◣	212	C LC_122	396.9	0.2	-4.3	-198.2	-114.9	98.1	78.0	OK	OK
		◢ 4.2 ◣	213	C LC_119	397.4	0.5	-196.4	140.0	142.0	98.2	65.4	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 7.0 ◣	69	C LC_135	230.8	0.0	169.7	84.6	31.7	57.0	45.0	OK	OK
		◢ 7.0 ◣	69	C LC_122	356.3	0.0	-36.1	-49.0	-198.7	88.0	51.9	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 4.2 ◣	212	C LC_119	398.7	1.3	-3.2	224.3	51.5	98.5	73.2	OK	OK
		◢ 4.2 ◣	212	C LC_119	398.7	1.3	17.5	215.6	80.1	98.5	76.3	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN56 side)	◢ 7.0 ◣	144	C LC_104	124.7	0.0	13.6	0.4	-71.6	30.8	19.8	OK	OK
		◢ 7.0 ◣	144	C LC_122	94.0	0.0	13.3	47.9	24.2	23.2	18.0	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 7.0 ◣	69	C LC_135	117.0	0.0	18.1	2.0	66.7	28.9	19.3	OK	OK
		◢ 7.0 ◣	68	C LC_122	104.6	0.0	34.3	-24.6	51.5	25.9	20.5	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 4.2 ◣	213	C LC_119	325.2	0.1	-301.6	-69.3	11.1	97.4	59.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	213	C LC_120	364.1	0.0	59.4	179.2	104.4	90.0	54.4	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 7.0 ◣	69	C LC_120	99.2	0.0	-28.6	17.5	-51.9	24.5	20.9	OK	OK
		◢ 7.0 ◣	68	C LC_122	95.2	0.0	37.9	12.3	48.9	23.5	18.7	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 4.2 ◣	212	C LC_119	396.8	0.1	-193.8	-74.0	-185.7	98.0	67.4	OK	OK
		◢ 4.2 ◣	213	C LC_119	349.5	0.1	301.7	38.2	-94.5	97.4	52.9	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 x 295 plate on ring bottom flange (SLN55 side)	◢ 7.0	311	C LC_119	397.0	0.2	-192.1	-4.9	-200.5	98.1	76.2	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 7.0 ◣	147	C LC_122	397.3	0.4	139.5	11.5	214.5	98.2	82.5	OK	OK
		◢ 7.0 ◣	147	C LC_119	396.7	0.0	-200.1	-133.1	-146.3	98.0	75.6	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 7.0 ◣	147	C LC_120	396.8	0.1	-122.0	-60.0	209.6	98.1	69.7	OK	OK
		◢ 7.0 ◣	146	C LC_122	396.7	0.1	-194.3	138.5	-143.9	98.0	71.1	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 x 295 plate on ring bottom flange (SLN56 side)	◢ 7.0	310	C LC_119	399.8	1.9	-79.4	72.5	214.3	98.8	87.7	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 x 295 plate on ring bottom flange (SLN56 side)	◢ 7.0	286	C LC_135	396.6	0.0	142.0	-156.0	146.3	98.0	75.3	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	287	C LC_153	271.3	0.0	26.1	-79.6	134.0	67.0	39.0	OK	OK
		◢ 4.2 ◣	288	C LC_120	236.4	0.0	-15.4	25.3	-133.8	58.4	32.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.1	0.3	-76.2	167.9	-149.9	98.1	58.7	OK	OK
		◢ 7.0 ◣	67	C LC_122	397.4	0.5	-117.0	-185.6	116.7	98.2	65.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.3	0.4	70.6	-177.3	-139.7	98.2	78.9	OK	OK
		◢ 7.0 ◣	67	C LC_122	397.4	0.5	123.9	190.1	106.7	98.2	74.9	OK	OK
395x145x20 fin plate upper (SLN55 side)	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	167	C LC_122	375.1	0.0	-218.9	-175.8	5.9	92.7	54.7	OK	OK
		◢ 4.2 ◣	166	C LC_122	396.7	0.0	148.0	-179.5	113.7	98.0	72.3	OK	OK
395x145x20 fin plate upper (SLN56 side)	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	167	C LC_135	398.3	1.0	100.9	-176.3	-135.6	98.4	72.8	OK	OK
		◢ 4.2 ◣	167	C LC_135	396.9	0.2	-267.3	41.7	-164.2	98.1	62.0	OK	OK
20 ring stiffener at 315 (SLN55 side)	20 x 295 plate on ring bottom flange (SLN55 side)	◢ 4.2 ◣	139	C LC_120	397.4	0.5	242.7	155.4	94.2	98.4	82.6	OK	OK
		◢ 4.2 ◣	139	C LC_120	403.0	3.9	1.5	209.1	102.1	99.6	95.3	OK	OK
20 bracket top plate a (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 7.0 ◣	67	C LC_122	167.5	0.0	-156.3	32.9	11.1	50.5	26.7	OK	OK
		◢ 7.0 ◣	67	C LC_122	230.8	0.0	157.9	-96.9	-7.2	57.0	35.9	OK	OK
20 bracket top plate b (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	67	C LC_122	206.0	0.0	-160.2	-74.8	-0.9	51.7	41.2	OK	OK
		◢ 7.0 ◣	67	C LC_122	255.5	0.0	-174.7	106.7	14.2	63.1	45.1	OK	OK
20 ring stiffener at 315 (SLN56 side)	20 x 295 plate on ring bottom flange (SLN56 side)	◢ 4.2 ◣	139	C LC_122	396.8	0.1	199.8	-179.7	-83.1	98.0	82.5	OK	OK
		◢ 4.2 ◣	139	C LC_122	399.3	1.7	246.0	-181.5	-6.2	98.7	93.4	OK	OK
20 bracket top plate a (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_122	363.9	0.0	-1.7	-49.2	-204.2	89.9	60.7	OK	OK
		◢ 7.0 ◣	67	C LC_122	342.6	0.0	-85.4	54.4	-183.7	84.7	57.6	OK	OK
20 bracket top plate b (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_122	396.6	0.0	-34.2	8.6	-228.0	98.0	68.9	OK	OK
		◢ 7.0 ◣	67	C LC_122	357.1	0.0	-3.7	-4.1	-206.1	88.2	60.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN55 side)	◢ 7.0 ◣	141	C LC_122	329.4	0.0	-105.5	-78.2	-162.3	81.4	53.5	OK	OK
		◢ 7.0 ◣	141	C LC_122	396.6	0.0	-44.4	-68.8	216.9	98.0	52.9	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 7.0 ◣	65	C LC_135	275.7	0.0	-123.8	-100.5	100.6	68.1	40.4	OK	OK
		◢ 7.0 ◣	65	C LC_135	171.6	0.0	-52.5	75.7	-56.3	42.4	28.8	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 7.0 ◣	37	C LC_120	402.4	3.6	-175.4	-105.8	-180.3	99.4	97.0	OK	OK
		◢ 7.0 ◣	37	C LC_120	399.8	2.0	-14.7	134.3	187.5	98.8	93.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 7.0 ◣	37	C LC_122	411.2	9.0	-64.6	-183.9	-145.5	101.6	100.6	OK	**Not OK!**
		◢ 7.0 ◣	37	C LC_104	406.0	5.8	-307.3	149.4	-33.7	100.3	100.0	OK	**Not OK!**
ring SLN55 + SLN56 (continuous)-bfl 1	20 x 295 plate on ring bottom flange (SLN55 side)	◢ 7.0	5	C LC_119	409.7	8.1	30.8	-52.9	-229.9	101.2	101.2	OK	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	189	C LC_122	397.1	0.3	-197.9	-95.5	-174.3	98.1	71.8	OK	OK
		◢ 4.2 ◣	189	C LC_122	378.9	0.0	66.7	214.3	21.2	93.6	55.9	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	39	C LC_122	398.9	1.4	271.0	-155.2	67.0	98.6	88.7	OK	OK
		◢ 7.0 ◣	39	C LC_122	399.5	1.8	107.9	222.0	5.6	98.7	90.3	OK	OK
20 x 295 plate on ring bottom flange (SLN55 side)	395x145x20 fin plate lower outer (SLN55 side)	◢ 4.2 ◣	67	C LC_119	416.1	12.0	155.9	-217.2	-49.1	102.8	100.4	OK	**Not OK!**
		◢ 4.2 ◣	67	C LC_119	402.1	3.4	-271.3	15.5	-170.6	101.9	98.2	OK	**Not OK!**
20 x 295 plate on ring bottom flange (SLN55 side)	395x145x20 fin plate lower inner (SLN55 side)	◢ 4.2 ◣	67	C LC_119	397.6	0.6	-216.6	172.7	85.0	98.2	70.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	67	C LC_119	401.1	2.8	270.9	-167.3	-34.1	99.1	87.4	OK	OK
20 x 295 plate on ring bottom flange (SLN56 side)	395x145x20 fin plate lower outer (SLN56 side)	◢ 4.2 ◣	67	C LC_119	398.4	1.0	-303.2	32.3	145.7	99.6	88.5	OK	OK
		◢ 4.2 ◣	67	C LC_119	407.2	6.5	53.3	224.2	63.8	100.6	100.0	OK	**Not OK!**
20 x 295 plate on ring bottom flange (SLN56 side)	395x145x20 fin plate lower inner (SLN56 side)	◢ 4.2 ◣	67	C LC_153	401.4	3.0	227.0	189.8	-22.6	99.2	92.3	OK	OK
		◢ 4.2 ◣	67	C LC_119	397.0	0.3	-85.5	-187.1	-123.0	98.1	82.3	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 4.2 ◣	222	C LC_122	201.3	0.0	-145.0	-75.8	27.6	49.8	39.5	OK	OK
		◢ 4.2 ◣	222	C LC_122	160.5	0.0	83.8	-4.8	78.9	39.7	31.8	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 4.2 ◣	222	C LC_122	269.7	0.0	-179.1	-39.6	-109.5	66.6	44.9	OK	OK
		◢ 4.2 ◣	222	C LC_122	249.3	0.0	120.1	-10.9	125.7	61.6	42.4	OK	OK
10 bracket web plate (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 4.2 ◣	222	C LC_122	396.6	0.0	23.7	10.6	228.3	98.0	74.6	OK	OK
		◢ 4.2 ◣	222	C LC_122	331.7	0.0	123.8	-174.7	-32.0	82.0	53.5	OK	OK
10 bracket web plate (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 4.2 ◣	222	C LC_135	396.6	0.0	2.9	3.3	229.0	98.0	65.3	OK	OK
		◢ 4.2 ◣	222	C LC_135	396.8	0.1	-23.7	218.0	-69.0	98.0	62.3	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate a (SLN55 side)	◢ 4.2 ◣	65	C LC_122	397.0	0.3	-137.4	-214.4	-17.2	98.1	76.4	OK	OK
		◢ 4.2 ◣	65	C LC_122	397.0	0.3	-66.0	-212.1	-78.1	98.1	72.9	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate b (SLN55 side)	◢ 4.2 ◣	68	C LC_122	400.2	2.2	-158.9	211.7	12.8	98.9	94.0	OK	OK
		◢ 4.2 ◣	68	C LC_122	400.0	2.1	-53.7	223.3	-50.2	98.8	90.7	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate a (SLN56 side)	◢ 4.2 ◣	68	C LC_135	398.0	0.9	-247.4	177.4	30.4	98.3	86.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	68	C LC_135	397.7	0.7	-150.7	206.7	49.2	98.3	83.5	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate b (SLN56 side)	◢ 4.2 ◣	65	C LC_122	397.2	0.3	-178.7	-203.3	24.4	98.1	73.5	OK	OK
		◢ 4.2 ◣	65	C LC_122	397.1	0.3	109.7	-207.6	74.0	98.1	73.7	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7

Item	Value	Unit	Reference
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5